# Predicting Medical Insurance Costs: The Role of Smoking, BMI, and Demographic Factors

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## 1 Introduction

Understanding the drivers of medical insurance costs is important for insurers, healthcare providers, and policymakers, as these costs vary widely across individuals due to differences in lifestyle, demographics, and health conditions. This project analyzes the insurance dataset from the book *Machine Learning with R* by Brett Lantz, which contains demographic and health-related variables associated with individual medical insurance charges. The dataset includes information on age, sex, body mass index (BMI), number of children, smoking status, geographic region, and annual insurance charges.

## 2 Problem

The goal of this project is to investigate how well demographic and health-related variables can predict medical insurance charges and to identify which factors most strongly influence variation in healthcare costs. A natural starting point is to fit a multiple linear regression model using age, BMI, number of children, smoking status, sex, and geographic region as predictors of insurance charges.

## 3 Data Preparation and Modeling Approach

### 3.1 Data Preparation

1. **Data loading and preprocessing.** The insurance dataset was imported into Python. Categorical variables were encoded for use in the models. Smoking status was converted to a binary indicator (1 = smoker, 0 = non-smoker), sex was converted to a binary indicator (1 = female, 0 = male), and geographic region was represented using dummy variables.

2. **Creation of derived variables.** For the Random Forest model, a binary outcome variable was created to indicate whether insurance charges were above or below the median. Observations with charges greater than or equal to the median insurance charge were labeled as high cost, while those below the median were labeled as low cost.

3. **Exploratory data analysis.** These analyses indicated that smoking status, age, and BMI were strongly associated with higher insurance costs.

### 3.2 Model Specifications

Using insurance charges as the primary outcome variable, several regression models were estimated to evaluate the effects of demographic and health-related variables on medical costs.

**Baseline linear regression model.** The baseline model uses ordinary least squares (OLS) regression with charges as the response variable and the predictors age, BMI, number of children, sex, smoking status, and geographic region:

$$charges \sim age + bmi + children + sex + smoker + region.$$

**Interaction models.** To examine whether certain risk factors compound one another, interaction terms were introduced into the regression models. The first interaction model evaluates whether the effect of BMI differs between smokers and non-smokers:

$$charges \sim age + bmi + children + sex + smoker + region + (bmi \times smoker).$$

The second interaction model tests whether the effect of age on insurance charges varies with smoking status:

$$charges \sim age + bmi + children + sex + smoker + region + (age \times smoker).$$

**Random Forest classification model.** To capture nonlinear relationships and complex predictor interactions, we also applied a Random Forest model. The response variable was defined as a binary indicator of whether an individual's insurance charges fall above or below the median cost. Model performance was evaluated using ROC-AUC, the F1 score, and the confusion matrix.

### 3.3 Model Diagnostics and Evaluation

For the linear regression models, residual versus fitted plots and Q–Q plots were examined to assess linearity, homoscedasticity, and residual normality. Because these assumptions were violated, bootstrap standard errors were computed to obtain more robust estimates of coefficient uncertainty.

For the Random Forest model, predictive performance was evaluated using ROC curves, AUC, the F1 score, and the confusion matrix.

We now evaluate how well demographic and lifestyle variables explain variation in insurance charges.

## 4 Data Description

The following table shows the variables included in the dataset, along with their units of measurement and variable type.

| Name of Variable | What does it measure? | Type of Variable | Units |
|---|---|---|---|
| Age | Age of the Primary Beneficiary | Nonnegative integer | Years |
| Sex | Sex of the insurance contractor | Binary (in this dataset), either male or female | n/a |
| BMI | Body-mass index of the beneficiary. | Continuous | kg/m^2 |
| Children | Number of children/dependents the beneficiary has | Nonnegative integer | n/a |
| Smoker | Whether or not the beneficiary smokes | Binary (yes/no) | n/a |
| Region | Region of the US the beneficiary lives in | Categorical (southeast, southwest, northeast, northwest) | n/a |
| Charges | Costs billed by medical insurance | Continuous | Dollars |

Figure 1: Description of Variables

Graphing the distributions of these variables gives us some more information:

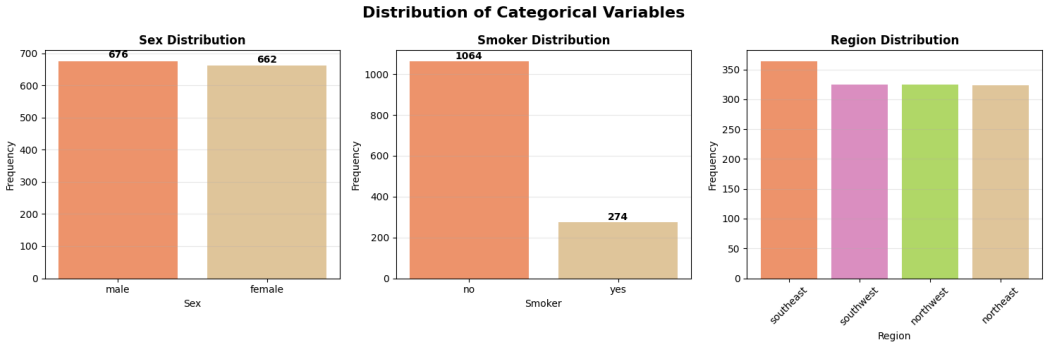


Figure 2: Distribution of Categorical Variables

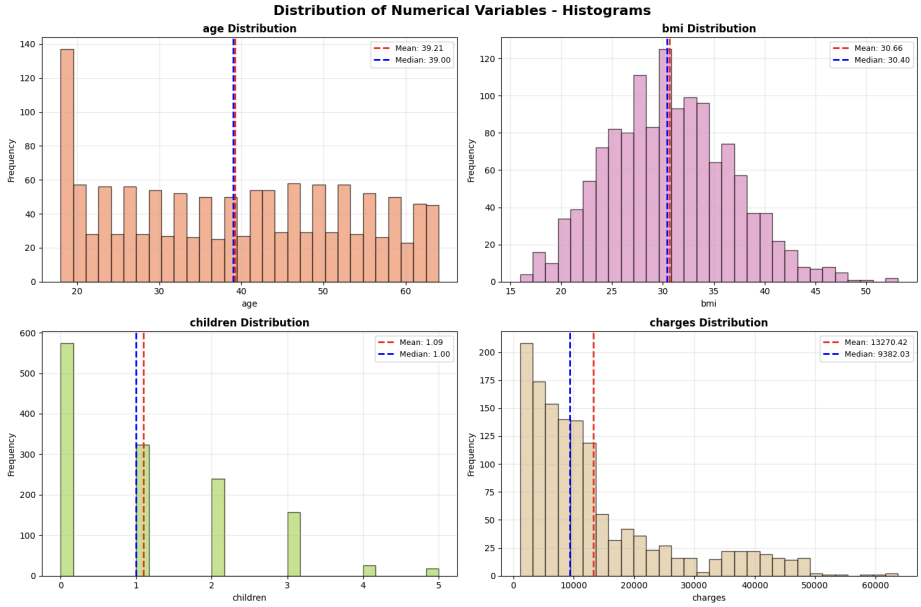


Figure 3: Distribution of Numerical Variables

We see that within the categorical variables (Figure 2), sex is well balanced, with male and female costs nearly equal. Smoking has a clear imbalance, with most individuals being non-smokers. Region was also fairly balanced with no large difference between region counts.

Within the numerical variables (Figure 3), age is approximately uniformly distributed across the range. BMI is roughly bell-shaped with mild right skew. Charges are strongly right-skewed, with a small number of high-cost outliers.

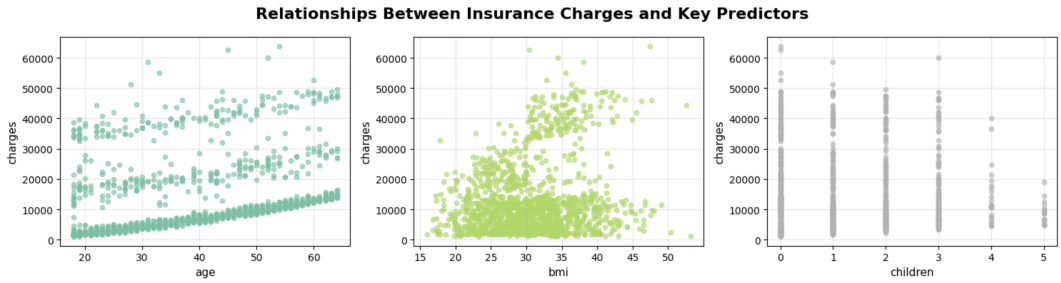


Figure 4: Relationships between insurance charges and key numerical predictors: age, BMI, and number of children.

Figure 4 presents scatter plots of insurance charges against the numerical predictors: age, BMI, and number of children. Charges increase substantially with age, indicating a strong positive relationship between age and medical costs. The relationship between BMI and charges is weaker but suggests that higher BMI values are associated with higher insurance costs, particularly among the largest charges. In contrast, the number of children shows little systematic relationship with charges and appears as discrete vertical bands due to the variable's integer nature. A full pairwise scatter matrix of the numerical variables is provided in Appendix A for completeness.

# 5 Data Analysis

Because charges are strongly right-skewed, we compute Spearman rank correlations to assess monotonic relationships between variables.

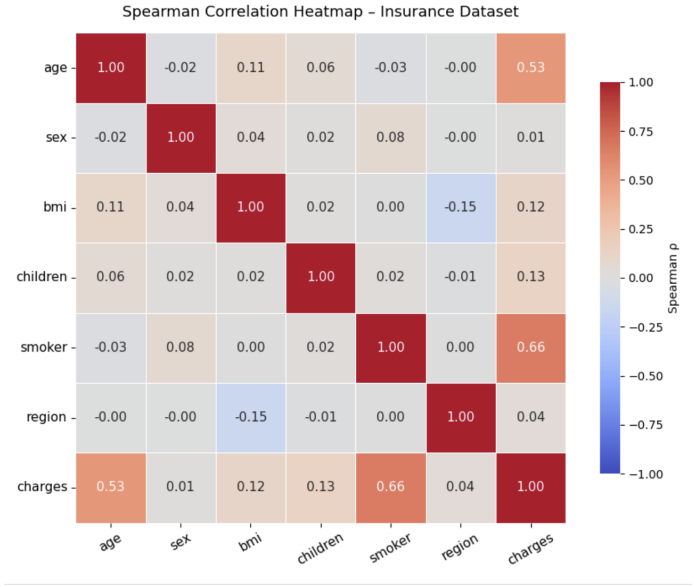


Figure 5: Spearman Correlation Matrix of Variables

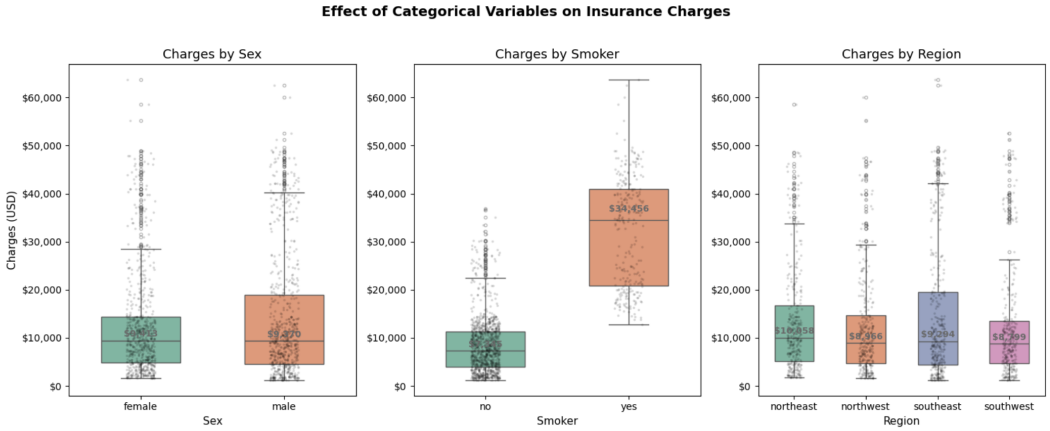


Figure 6: Correlation of Categorical Variables with Charges

The correlation analysis highlights smoking status as the variable most strongly associated with insurance charges. Smokers incur substantially higher costs than non-smokers. Stratifying BMI by smoking status reveals that high BMI (above 30) is associated with increased charges primarily among smokers, suggesting an interaction between BMI and smoking.

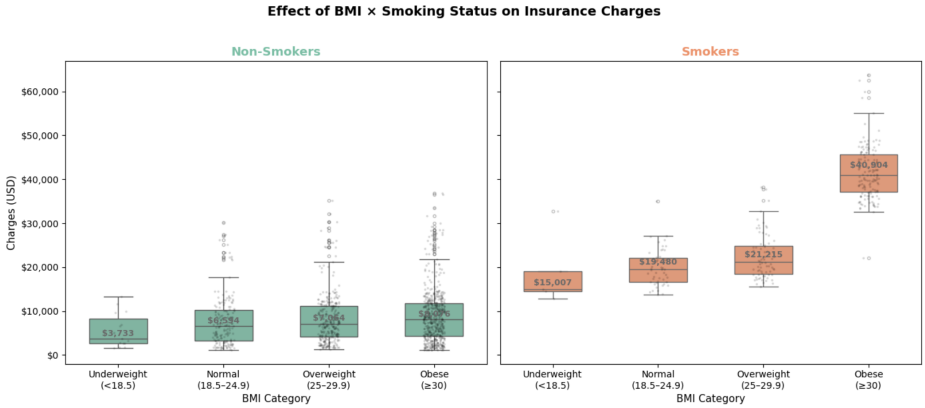


Figure 7: Correlation of BMI with Charges for Smokers vs. Non-smokers

Following the correlation analysis, we estimate a multiple linear regression model to examine how age, BMI, number of children, sex, smoking status, and region predict insurance charges.

OLS Regression Summary Table

| | Variable | Coefficient | Std. Error | t-value | p-value |
|---|---|---|---|---|---|
| 0 | Intercept | -11938.54 | 987.82 | -12.09 | 0.000 |
| 1 | Male | -131.31 | 332.95 | -0.39 | 0.693 |
| 2 | Smoker | 23848.53 | 413.15 | 57.72 | 0.000 |
| 3 | Region: Northwest | -352.96 | 476.28 | -0.74 | 0.459 |
| 4 | Region: Southeast | -1035.02 | 478.69 | -2.16 | 0.031 |
| 5 | Region: Southwest | -960.05 | 477.93 | -2.01 | 0.045 |
| 6 | Age | 256.86 | 11.90 | 21.59 | 0.000 |
| 7 | BMI | 339.19 | 28.60 | 11.86 | 0.000 |
| 8 | Children | 475.50 | 137.80 | 3.45 | 0.001 |

Model Fit Statistics
R-squared: 0.751
Adjusted R-squared: 0.749
Observations: 1338

Figure 8: OLS regression summary table for predicting insurance charges using demographic and health predictors.

Figure 8 summarizes the results of the baseline multiple linear regression predicting insurance charges using demographic and health predictors. Smoking status emerges as the strongest predictor, with smokers incurring approximately \$23,850 higher charges on average than non-smokers, holding other variables constant. Age and BMI also show significant positive relationships with costs, increasing charges by about \$257 per year of age and \$339 per BMI unit. Although statistically significant, the magnitude of the children coefficient is relatively small compared to smoking or BMI, and region effects are statistically weak and economically small relative to smoking, BMI, and age.

The model explains a substantial portion of the variation in charges ($R^2 = 0.751$). The full regression output and additional diagnostic statistics are provided in Appendix A.

To assess model assumptions, we examined residual versus fitted plots and Q–Q plots to evaluate linearity, heteroskedasticity, and residual normality.

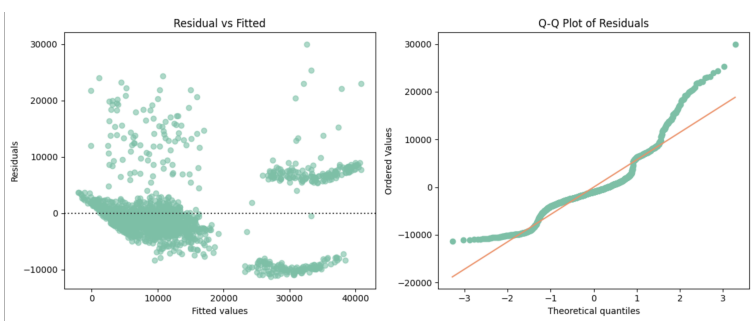


Figure 9: Diagnostic plots for the OLS regression model, including residual vs. fitted values and the Q-Q plot of residuals.

The residual versus fitted plot shows a clear pattern rather than random scatter, with the spread of residuals increasing at higher fitted values. This indicates heteroskedasticity, meaning the variance of residuals increases with predicted charges. Distinct clusters of residuals also appear at higher fitted values, likely reflecting differences between smokers and non-smokers.

The Q–Q plot shows deviations from the reference line, particularly in the upper tail, indicating that the

residuals are not normally distributed.

Formal Regression Assumption Tests

| | Test | Statistic | p-value | Conclusion |
|---|---|---|---|---|
| 0 | Breusch-Pagan (LM) | 121.7436 | <0.0001 | Heteroskedasticity present |
| 1 | Breusch-Pagan (F) | 16.6286 | <0.0001 | Heteroskedasticity present |
| 2 | Shapiro-Wilk | 0.8989 | <0.0001 | Residuals not normal |
| 3 | Durbin-Watson | 2.0884 | | No autocorrelation concern |

Variance Inflation Factors
(Multicollinearity Check)

| | Variable | VIF |
|---|---|---|
| 0 | Region: southeast | 1.652 |
| 1 | Region: southwest | 1.529 |
| 2 | Region: northwest | 1.519 |
| 3 | bmi | 1.107 |
| 4 | age | 1.017 |
| 5 | Smoker | 1.012 |
| 6 | Male | 1.009 |
| 7 | children | 1.004 |

Figure 10: Regression diagnostic tests and variance inflation factors (VIF) for the baseline OLS model.

The Breusch–Pagan tests strongly reject the null hypothesis of constant error variance ($p < 0.0001$), indicating heteroskedasticity in the regression residuals. The Shapiro–Wilk test also rejects normality of residuals ($p < 0.0001$), consistent with the right-skewed distribution of insurance charges. In contrast, the Durbin–Watson statistic is close to 2, suggesting no evidence of autocorrelation. Variance inflation factors are all close to 1 and well below common thresholds for problematic multicollinearity, indicating that the predictors are not strongly correlated. Because heteroskedasticity and non-normality are detected in the residuals, bootstrap standard errors are used to provide more robust inference for the regression coefficients. As an additional robustness check for the right-skewed cost distribution, we also estimated a Gamma generalized linear model with a log link. The results, reported in Appendix A, show similar coefficient patterns and a lower AIC than the OLS model, suggesting improved fit for skewed cost data.

SE Comparison (OLS vs Bootstrap):

| | Coefficient | OLS SE | Bootstrap SE |
|---|---|---|---|
| 0 | Intercept | 987.819175 | 1032.045761 |
| 1 | sex[T.male] | 332.945439 | 325.805706 |
| 2 | smoker[T.yes] | 413.153355 | 579.782288 |
| 3 | region[T.northwest] | 476.275786 | 479.359903 |
| 4 | region[T.southeast] | 478.692209 | 508.170021 |
| 5 | region[T.southwest] | 477.933024 | 459.674496 |
| 6 | age | 11.898849 | 11.338679 |
| 7 | bmi | 28.599470 | 31.652729 |
| 8 | children | 137.804093 | 132.781179 |

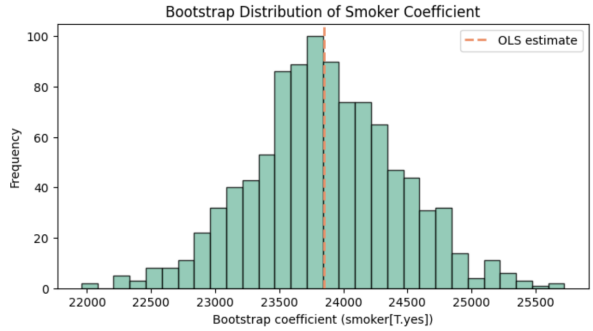


Figure 11: Comparison of OLS and bootstrap standard errors along with the bootstrap distribution of the smoking coefficient. The dashed line indicates the OLS estimate.

The table above compares the classical OLS standard errors with the bootstrap estimates. For several coefficients, particularly the smoking variable, the bootstrap standard errors are slightly larger than the OLS estimates, suggesting that the classical regression may underestimate uncertainty when model assumptions are violated.

The figure also shows the bootstrap distribution of the smoking coefficient. The distribution is approximately centered around the OLS estimate, indicating that the estimated effect of smoking on insurance charges is stable across resampled datasets. This result provides additional evidence that smoking has a

strong and consistent impact on medical insurance costs.

## 5.1 Interaction Effects

To investigate whether key predictors jointly influence medical charges, we extend the baseline regression model by incorporating interaction terms between important demographic and health variables. Interaction models allow the effect of one predictor to vary depending on the value of another predictor, capturing potential compounding risk factors that may not be visible in an additive model.

We first examine whether the relationship between BMI and insurance charges differs between smokers and non-smokers. The interaction term between BMI and smoking status is statistically significant, indicating that the effect of BMI on medical costs depends on smoking behavior. Additional regression diagnostics are reported in the Appendix.

The figure below illustrates this interaction. Smokers exhibit a much steeper increase in charges as BMI rises than non-smokers. This suggests that a higher BMI amplifies the financial impact of smoking on medical expenses, reflecting compounding health risks associated with obesity and smoking simultaneously.

BMI × Smoker Interaction Model Summary

| | Variable | Coefficient | Std. Error | t-value | p-value |
|---|---|---|---|---|---|
| 0 | Intercept | -2223.45 | 865.61 | -2.57 | 0.010 |
| 1 | Male | -500.15 | 266.52 | -1.88 | 0.061 |
| 2 | Smoker | -20415.61 | 1648.28 | -12.39 | 0.000 |
| 3 | Region: Northwest | -585.48 | 380.86 | -1.54 | 0.124 |
| 4 | Region: Southeast | -1210.13 | 382.75 | -3.16 | 0.002 |
| 5 | Region: Southwest | -1231.11 | 382.22 | -3.22 | 0.001 |
| 6 | Age | 263.62 | 9.52 | 27.70 | 0.000 |
| 7 | BMI | 23.53 | 25.60 | 0.92 | 0.358 |
| 8 | BMI × Smoker | 1443.10 | 52.65 | 27.41 | 0.000 |
| 9 | Children | 516.40 | 110.18 | 4.69 | 0.000 |

Model Fit Statistics
R-squared: 0.841
Adjusted R-squared: 0.84
Observations: 1338

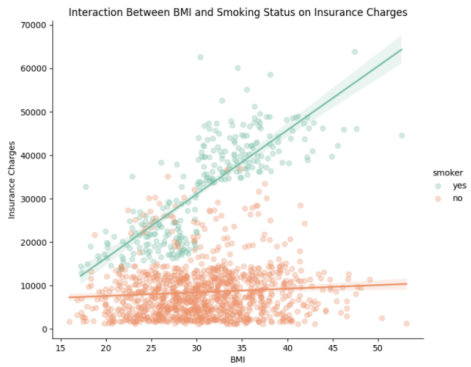


Figure 12: Interaction between BMI and smoking status on insurance charges.

We also test whether the effect of age on insurance charges differs by smoking status. The interaction model suggests that medical costs increase with age for both groups, but the increase is substantially larger for smokers.

Age × Smoker Interaction Model Summary

| | Variable | Coefficient | Std. Error | t-value | p-value |
|---|---|---|---|---|---|
| 0 | Intercept | -11612.13 | 1010.15 | -11.50 | 0.000 |
| 1 | Male | -135.16 | 332.79 | -0.41 | 0.685 |
| 2 | Smoker | 22105.00 | 1213.13 | 18.22 | 0.000 |
| 3 | Region: Northwest | -362.54 | 476.08 | -0.76 | 0.446 |
| 4 | Region: Southeast | -1060.06 | 478.73 | -2.21 | 0.027 |
| 5 | Region: Southwest | -943.31 | 477.82 | -1.97 | 0.049 |
| 6 | Age | 247.73 | 13.31 | 18.62 | 0.000 |
| 7 | Age × Smoker | 45.13 | 29.53 | 1.53 | 0.127 |
| 8 | BMI | 340.62 | 28.60 | 11.91 | 0.000 |
| 9 | Children | 471.41 | 137.76 | 3.42 | 0.001 |

Model Fit Statistics
R-squared: 0.751
Adjusted R-squared: 0.75
Observations: 1338

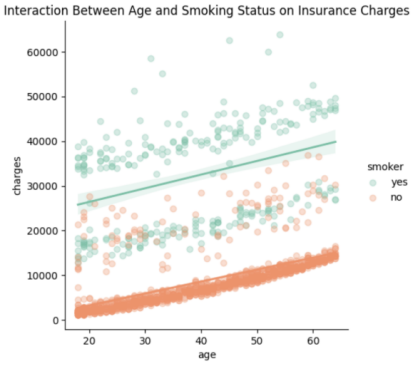


Figure 13: Interaction between age and smoking status on insurance charges.

Model Comparison Table

| | Model | R² | Adj R² | AIC | BIC | RMSE | BP p-value | Shapiro p-value | Durbin-Watson |
|---|---|---|---|---|---|---|---|---|---|
| 0 | BMI × Smoker | 0.841 | 0.840 | 26515.6 | 26567.6 | 4828.28 | 0.3427 | 0.0 | 2.069 |
| 1 | Age × Smoker | 0.751 | 0.750 | 27113.2 | 27165.1 | 6036.37 | 0.0000 | 0.0 | 2.093 |
| 2 | Base OLS | 0.751 | 0.749 | 27113.5 | 27160.3 | 6041.68 | 0.0000 | 0.0 | 2.088 |
| 3 | Age × BMI | 0.751 | 0.749 | 27115.5 | 27167.5 | 6041.65 | 0.0000 | 0.0 | 2.088 |

Figure 14: Comparison of regression models predicting insurance charges.

The model comparison results indicate that the BMI × smoker interaction model provides the best overall fit. This specification achieves the highest explanatory power ($R^2 = 0.841$) and the lowest prediction error (RMSE = 4828), substantially improving upon the baseline OLS model ($R^2 = 0.751$). Information criteria (AIC and BIC) also favor this model, and the Breusch–Pagan test suggests that heteroskedasticity is reduced under this specification. These results indicate that the interaction between smoking status and BMI captures an important interaction effect in insurance charges.

However, diagnostic analyses still suggest the presence of nonlinear relationships and residual variance patterns that may not be fully captured by linear regression. While regression models help explain how demographic and health-related variables influence the magnitude of insurance charges, insurers are often particularly interested in identifying individuals who are likely to incur unusually high costs. To examine this predictive perspective, we also frame the problem as a classification task by labeling observations as high-cost or low-cost based on whether their charges exceed the median.

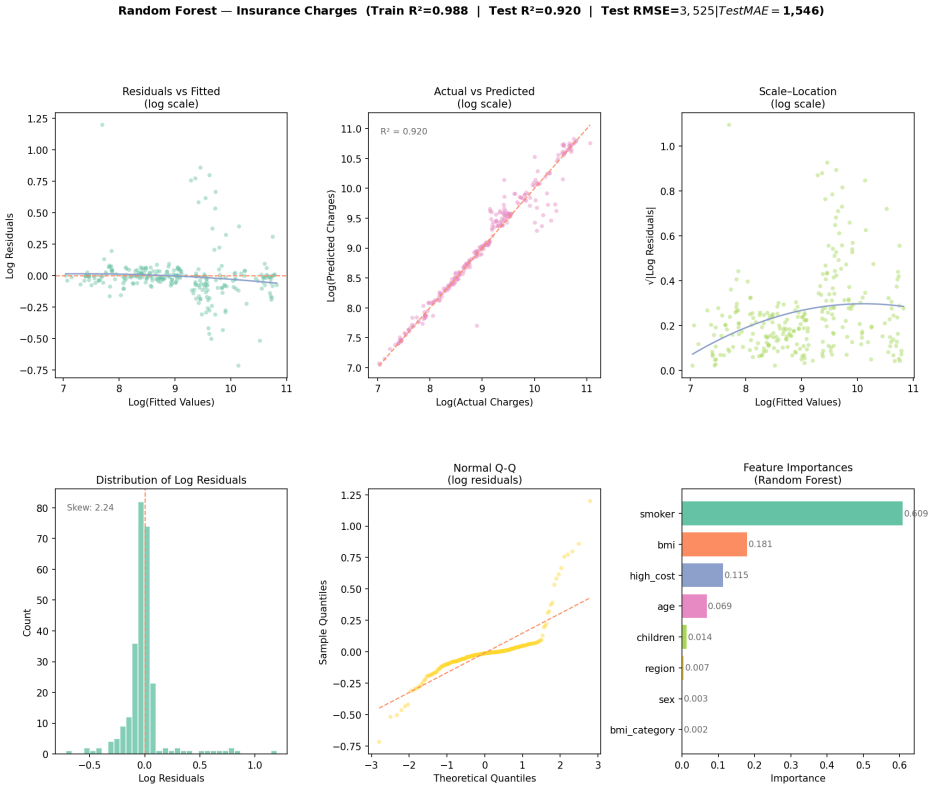


Figure 15: Random Forest diagnostic plots for insurance charge prediction.

The Random Forest model demonstrates strong predictive performance for insurance charges, achieving a test $R^2$ of approximately 0.92. The actual versus predicted plot shows that most observations lie close to the 45-degree reference line, indicating accurate predictions across the range of charges. Residual plots on

the log scale show reduced systematic patterns compared to the linear regression diagnostics, suggesting the model captures nonlinear relationships more effectively. The feature importance analysis further confirms that smoking status is the most influential predictor of insurance costs, followed by BMI and age, reinforcing the key findings from the earlier regression analysis.

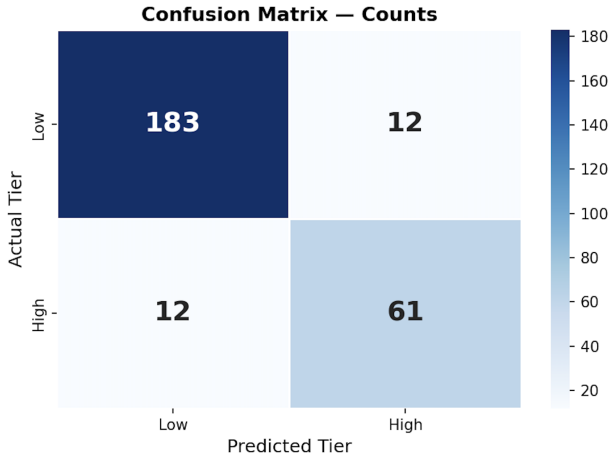


Figure 16: Confusion Matrix for High-cost vs. Low-cost Charges

To evaluate the Random Forest model's ability to distinguish high-cost from low-cost patients, we split predicted charges into two tiers, with low-cost charges corresponding to the 0-70th percentiles and high-cost charges to the 71-100th percentile.

We apply a train-test split to the data, with 20% of the data classified as a test dataset. We then apply the Random Forest model to the test dataset (n=268). From the ROC model (shown in the appendix), we see that the model has a strongly discriminatory predictive performance, with an AUC of 0.918. The confusion matrix corroborates this data, and we see that the model correctly classified 94% of low-cost individuals and 84% of high-cost individuals, with an overall accuracy of roughly 92%. This is better than one would expect under a standard OLS model or an OLS model + interactions.

# 6 Conclusions

We compared several models for predicting individual medical insurance charges using demographic and health-related variables. The baseline OLS regression explained a substantial portion of the variation in charges ($R^2 = 0.751$) and identified smoking status as the dominant predictor, followed by BMI and age. However, diagnostic tests revealed violations of key regression assumptions, including heteroskedasticity and non-normal residuals. Bootstrap standard errors were therefore used to provide more reliable inference.

Interaction models further improved model performance. In particular, the BMI $\times$ smoker specification substantially increased explanatory power ($R^2 = 0.841$) and reduced prediction error relative to the baseline model. These results indicate that smoking amplifies the impact of other health-related risk factors, especially BMI and age, revealing important compounding effects in medical insurance costs.

A Random Forest model was also estimated to capture nonlinear relationships and complex interactions among predictors. This model achieved strong predictive performance with a test $R^2$ of approximately 0.92. While the Random Forest provides the highest predictive accuracy, its black-box structure limits interpretability relative to regression-based approaches.

Given the goal of understanding and predicting medical insurance charges, the regression model including the BMI $\times$ smoker interaction provides the most appropriate balance between predictive performance and interpretability. It improves explanatory power relative to the baseline model while maintaining a transparent structure that allows the influence of key predictors to be clearly interpreted. Overall, the results highlight the substantial financial impact of lifestyle-related risk factors—particularly smoking and obesity—in determining healthcare expenditures.

# A Appendix

## A.1 Contribution & LLM Usage statement

Many of the results were taken from previous homework assignments, for which contribution statements were made separately. Sowmya and Rishika worked on both the code notebook and the written report together. LLM was used to verify correctness of results, edit LaTeX formatting, and provide pseudocode where needed.

## A.2 Additional Exploratory Data Analysis

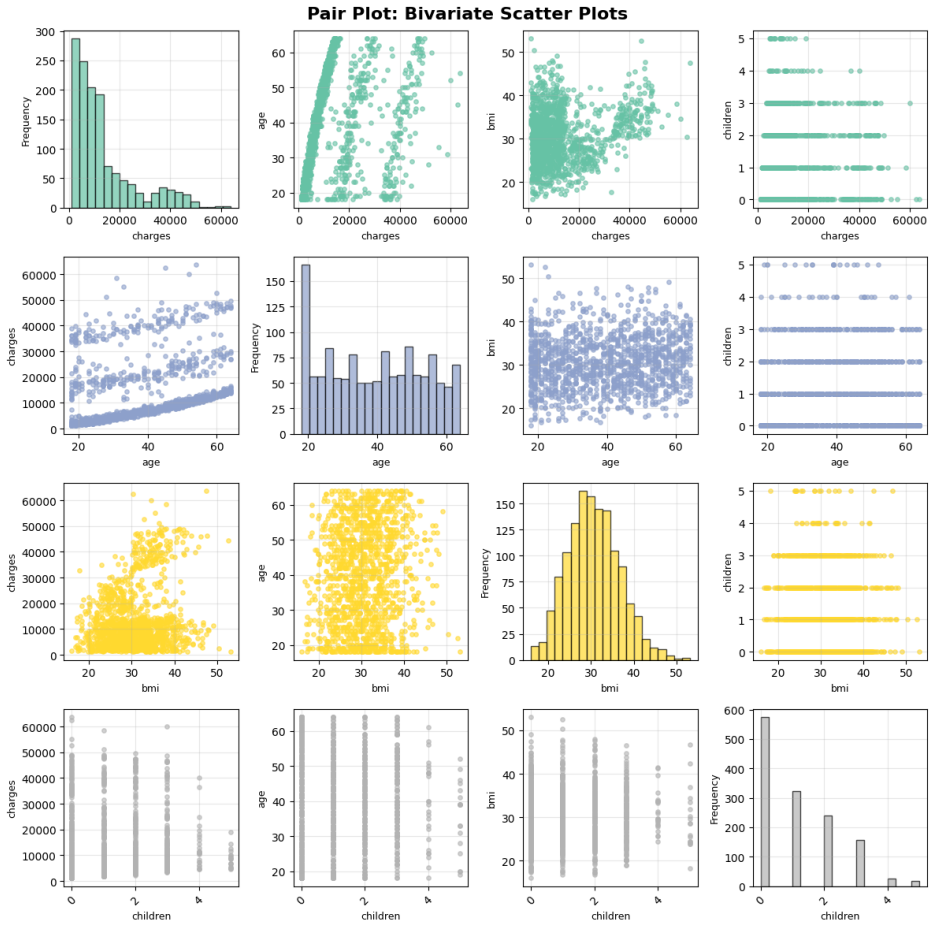


Figure 17: Full pairwise scatter matrix of numerical variables in the insurance dataset

## A.3 Full OLS Regression Outputs

The complete statsmodels regression output for all OLS models are shown below in figures 16-18.

OLS Regression Results

| | | | |
|---|---|---|---|
| **Dep. Variable:** | charges | **R-squared:** | 0.751 |
| **Model:** | OLS | **Adj. R-squared:** | 0.749 |
| **Method:** | Least Squares | **F-statistic:** | 500.8 |
| **Date:** | Fri, 13 Mar 2026 | **Prob (F-statistic):** | 0.00 |
| **Time:** | 03:20:13 | **Log-Likelihood:** | -13548. |
| **No. Observations:** | 1338 | **AIC:** | 2.711e+04 |
| **Df Residuals:** | 1329 | **BIC:** | 2.716e+04 |
| **Df Model:** | 8 | | |
| **Covariance Type:** | nonrobust | | |

| | **coef** | **std err** | **t** | **P>\|t\|** | **[0.025** | **0.975]** |
|---|---|---|---|---|---|---|
| **Intercept** | -1.194e+04 | 987.819 | -12.086 | 0.000 | -1.39e+04 | -1e+04 |
| **sex[T.male]** | -131.3144 | 332.945 | -0.394 | 0.693 | -784.470 | 521.842 |
| **smoker[T.yes]** | 2.385e+04 | 413.153 | 57.723 | 0.000 | 2.3e+04 | 2.47e+04 |
| **region[T.northwest]** | -352.9639 | 476.276 | -0.741 | 0.459 | -1287.298 | 581.370 |
| **region[T.southeast]** | -1035.0220 | 478.692 | -2.162 | 0.031 | -1974.097 | -95.947 |
| **region[T.southwest]** | -960.0510 | 477.933 | -2.009 | 0.045 | -1897.636 | -22.466 |
| **age** | 256.8564 | 11.899 | 21.587 | 0.000 | 233.514 | 280.199 |
| **bmi** | 339.1935 | 28.599 | 11.860 | 0.000 | 283.088 | 395.298 |
| **children** | 475.5005 | 137.804 | 3.451 | 0.001 | 205.163 | 745.838 |

| | | | |
|---|---|---|---|
| **Omnibus:** | 300.366 | **Durbin-Watson:** | 2.088 |
| **Prob(Omnibus):** | 0.000 | **Jarque-Bera (JB):** | 718.887 |
| **Skew:** | 1.211 | **Prob(JB):** | 7.86e-157 |
| **Kurtosis:** | 5.651 | **Cond. No.** | 311. |

Figure 18: OLS regression results for predicting insurance charges using demographic and health predictors.

OLS Regression Results

| | | | |
|---|---|---|---|
| **Dep. Variable:** | charges | **R-squared:** | 0.841 |
| **Model:** | OLS | **Adj. R-squared:** | 0.840 |
| **Method:** | Least Squares | **F-statistic:** | 780.0 |
| **Date:** | Fri, 13 Mar 2026 | **Prob (F-statistic):** | 0.00 |
| **Time:** | 04:26:35 | **Log-Likelihood:** | -13248. |
| **No. Observations:** | 1338 | **AIC:** | 2.652e+04 |
| **Df Residuals:** | 1328 | **BIC:** | 2.657e+04 |
| **Df Model:** | 9 | | |
| **Covariance Type:** | nonrobust | | |

| | **coef** | **std err** | **t** | **P>\|t\|** | **[0.025** | **0.975]** |
|---|---|---|---|---|---|---|
| **Intercept** | -2223.4539 | 865.611 | -2.569 | 0.010 | -3921.569 | -525.339 |
| **sex[T.male]** | -500.1460 | 266.518 | -1.877 | 0.061 | -1022.987 | 22.695 |
| **smoker[T.yes]** | -2.042e+04 | 1648.277 | -12.386 | 0.000 | -2.36e+04 | -1.72e+04 |
| **region[T.northwest]** | -585.4780 | 380.859 | -1.537 | 0.124 | -1332.630 | 161.674 |
| **region[T.southeast]** | -1210.1312 | 382.750 | -3.162 | 0.002 | -1960.992 | -459.270 |
| **region[T.southwest]** | -1231.1077 | 382.218 | -3.221 | 0.001 | -1980.924 | -481.291 |
| **age** | 263.6202 | 9.516 | 27.703 | 0.000 | 244.952 | 282.288 |
| **bmi** | 23.5329 | 25.601 | 0.919 | 0.358 | -26.689 | 73.755 |
| **bmi:smoker[T.yes]** | 1443.0964 | 52.647 | 27.411 | 0.000 | 1339.816 | 1546.377 |
| **children** | 516.4034 | 110.179 | 4.687 | 0.000 | 300.259 | 732.548 |

| | | | |
|---|---|---|---|
| **Omnibus:** | 721.596 | **Durbin-Watson:** | 2.069 |
| **Prob(Omnibus):** | 0.000 | **Jarque-Bera (JB):** | 4465.236 |
| **Skew:** | 2.529 | **Prob(JB):** | 0.00 |
| **Kurtosis:** | 10.384 | **Cond. No.** | 663. |

Figure 19: Full OLS Results for BMI and Smoking Interaction

OLS Regression Results

| | | | |
|---|---|---|---|
| **Dep. Variable:** | charges | **R-squared:** | 0.751 |
| **Model:** | OLS | **Adj. R-squared:** | 0.750 |
| **Method:** | Least Squares | **F-statistic:** | 445.9 |
| **Date:** | Fri, 13 Mar 2026 | **Prob (F-statistic):** | 0.00 |
| **Time:** | 04:25:02 | **Log-Likelihood:** | -13547. |
| **No. Observations:** | 1338 | **AIC:** | 2.711e+04 |
| **Df Residuals:** | 1328 | **BIC:** | 2.717e+04 |
| **Df Model:** | 9 | | |
| **Covariance Type:** | nonrobust | | |

| | **coef** | **std err** | **t** | **P>\|t\|** | **[0.025** | **0.975]** |
|---|---|---|---|---|---|---|
| **Intercept** | -1.161e+04 | 1010.153 | -11.495 | 0.000 | -1.36e+04 | -9630.463 |
| **sex[T.male]** | -135.1579 | 332.788 | -0.406 | 0.685 | -788.005 | 517.689 |
| **smoker[T.yes]** | 2.211e+04 | 1213.127 | 18.222 | 0.000 | 1.97e+04 | 2.45e+04 |
| **region[T.northwest]** | -362.5442 | 476.078 | -0.762 | 0.446 | -1296.491 | 571.402 |
| **region[T.southeast]** | -1060.0595 | 478.732 | -2.214 | 0.027 | -1999.213 | -120.906 |
| **region[T.southwest]** | -943.3076 | 477.819 | -1.974 | 0.049 | -1880.669 | -5.946 |
| **age** | 247.7328 | 13.307 | 18.617 | 0.000 | 221.628 | 273.837 |
| **age:smoker[T.yes]** | 45.1293 | 29.525 | 1.529 | 0.127 | -12.792 | 103.050 |
| **bmi** | 340.6250 | 28.600 | 11.910 | 0.000 | 284.518 | 396.732 |
| **children** | 471.4128 | 137.761 | 3.422 | 0.001 | 201.160 | 741.665 |

| | | | |
|---|---|---|---|
| **Omnibus:** | 300.122 | **Durbin-Watson:** | 2.093 |
| **Prob(Omnibus):** | 0.000 | **Jarque-Bera (JB):** | 717.194 |
| **Skew:** | 1.211 | **Prob(JB):** | 1.83e-156 |
| **Kurtosis:** | 5.646 | **Cond. No.** | 399. |

Figure 20: Full OLS Results for Age and Smoking Interaction

## A.4 Gamma GLM Robustness Check

Because insurance charges are strictly positive and strongly right-skewed, we also estimated a generalized linear model with a Gamma distribution and log link as a robustness check. Gamma GLMs are commonly used for modeling healthcare expenditures because they naturally accommodate skewed cost distributions and heteroskedasticity.

The estimated coefficients are broadly consistent with the OLS results reported in the main analysis. Smoking status remains the strongest predictor of insurance costs, while age and BMI also show statistically significant positive effects.

Model comparison using AIC indicates that the Gamma GLM provides a better overall fit than the OLS model (AIC = 26632.8 vs. 27113.5), suggesting that the Gamma specification more appropriately captures the distribution of insurance charges.

Generalized Linear Model Regression Results

| | | | |
|---|---|---|---|
| **Dep. Variable:** | y | **No. Observations:** | 1338 |
| **Model:** | GLM | **Df Residuals:** | 1329 |
| **Model Family:** | Gamma | **Df Model:** | 8 |
| **Link Function:** | log | **Scale:** | 0.46695 |
| **Method:** | IRLS | **Log-Likelihood:** | -13307. |
| **Date:** | Fri, 13 Mar 2026 | **Deviance:** | 337.73 |
| **Time:** | 20:09:19 | **Pearson chi2:** | 621. |
| **No. Iterations:** | 18 | **Pseudo R-squ. (CS):** | 0.6833 |
| **Covariance Type:** | nonrobust | | |

| | coef | std err | z | P>\|z\| | [0.025 | 0.975] |
|---|---|---|---|---|---|---|
| **Intercept** | 7.3863 | 0.111 | 66.334 | 0.000 | 7.168 | 7.605 |
| **sex[T.male]** | -0.0571 | 0.038 | -1.521 | 0.128 | -0.131 | 0.016 |
| **smoker[T.yes]** | 1.5003 | 0.047 | 32.215 | 0.000 | 1.409 | 1.592 |
| **region[T.northwest]** | -0.0579 | 0.054 | -1.078 | 0.281 | -0.163 | 0.047 |
| **region[T.southeast]** | -0.1416 | 0.054 | -2.625 | 0.009 | -0.247 | -0.036 |
| **region[T.southwest]** | -0.1455 | 0.054 | -2.700 | 0.007 | -0.251 | -0.040 |
| **age** | 0.0286 | 0.001 | 21.351 | 0.000 | 0.026 | 0.031 |
| **bmi** | 0.0141 | 0.003 | 4.381 | 0.000 | 0.008 | 0.020 |
| **children** | 0.0842 | 0.016 | 5.419 | 0.000 | 0.054 | 0.115 |

```
Model comparison (AIC):
  OLS:       27113.5
  Gamma GLM: 26632.8
```

Figure 21: Gamma generalized linear model with log link for predicting insurance charges.

## A.5 ROC curve

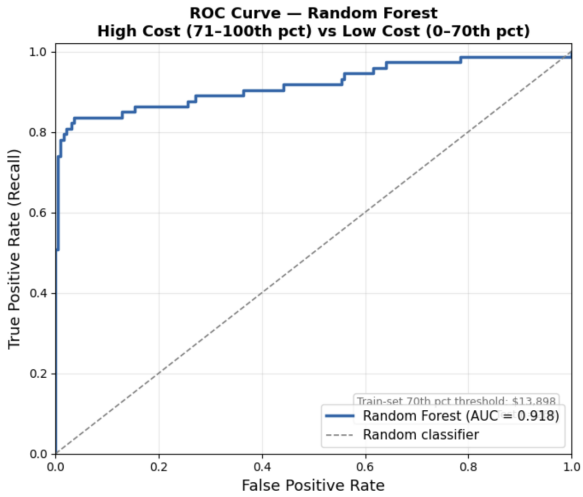


Figure 22: ROC curve of High-cost vs. Low-cost